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1. LINUX BASICS

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- Linux is a powerful, multi-user, multi-tasking operating system.
- Linux contains a kernel, which is the core part of the operating system.
- The kernel manages system resources and communication between hardware and software.

Terminal:

- A window where we run commands through a shell. The shell communicates with the computer.

Important commands:

`pwd`

Print working directory. Shows the current location.

`ls`

Lists files and directories.

`cd`

Changes directory.

Examples:

`cd folder`

`cd ..`

`cd ~`

~ represents the user's home directory.

mkdir

Creates a new directory.

rm

Removes files.

rmdir

Removes directories when empty.

cp

Copies files.

mv

Moves or renames files.

cat

Displays file contents.

less

Reads files one page at a time.

wc

Counts lines, words and characters.

Examples:

wc -w file.txt -> count words

wc -l file.txt -> count lines

ssh

Remote login to another computer/server.

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2. PATHS

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Full Path:

Starts from the root directory (/).

It is independent of your current location.

Example:

/home/user/texts/file.txt

Relative Path:

Starts from your current directory.

Special paths:

.

Current directory

..

One directory above

../..

Two directories above

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3. COMMAND OPTIONS

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Many Linux commands accept options.

Example:

`ls -l`

Shows detailed file information.

`ls -h`

Human readable sizes.

Options modify command behaviour.

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4. PIPELINES

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The pipe symbol `|` connects commands.

The output of the command on the left becomes the input of the command on the right.

Example:

`cat file.txt | wc -w`

First `cat` prints the file.

Then `wc` counts words.

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5. WORD FREQUENCY PIPELINE

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Command:

```
cat file.txt | egrep -o '\w+' | sort | uniq -c | sort -rn
```

Step 1:

cat reads file contents.

Step 2:

egrep -o '\w+' extracts individual words.

-o means only print matching parts.

\w+ means one or more word characters.

Step 3:

sort places identical words together.

Step 4:

uniq -c counts repeated words.

Step 5:

sort -rn sorts numerically in reverse order.

Result:

Most frequent words appear first.

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6. ENVIRONMENT VARIABLES

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Environment variables store system information.

Important variables:

\$PATH

Contains directories where the shell searches for executable programs.

\$HOME

Home directory path.

\$PWD

Current directory.

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7. PYTHON BASICS

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Python 3 runs the Python interpreter.

Running a file:

`python3 program.py`

Command line arguments:

```
import sys
```

```
sys.argv[0]
```

Contains script name.

```
sys.argv[1:]
```

Contains user supplied arguments.

Example:

```
python3 test.py hello
```

sys.argv contains:

```
['test.py', 'hello']
```

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8. REGULAR EXPRESSIONS

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grep and egrep search text patterns.

Example:

```
egrep 'apple|pear' file.txt
```

Finds lines containing apple OR pear.

Useful symbols:

*

Zero or more repetitions

+

One or more repetitions

?

Optional character

|

OR operation

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9. FILE EDITOR MICRO

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micro is a terminal text editor.

Useful shortcuts:

Ctrl + S

Save

Ctrl + Q

Quit

Ctrl + C

Copy

Ctrl + V

Paste

Ctrl + E

Command prompt

vsplit

Vertical split screen

hsplit

Horizontal split screen

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10. PRACTICE EXERCISES

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Solved concepts:

1. Explain ~

Answer:

~ represents the home directory.

2. Explain pipe |

Answer:

It sends output from one command as input to another.

3. Explain egrep -o '\w+'

Answer:

Extracts individual words from text.

4. Explain uniq -c

Answer:

Counts adjacent duplicate lines.